

AthenaK CGL–Landau-Fluid Validation and MKS24 Analysis Note

May 25, 2026

1 Purpose

This note integrates the illustrated validation report originally maintained on branch `CGL–STS–LF` into the current `feature/cgl-landau-fluid` implementation. It documents the AthenaK CGL–Landau-fluid (CGL–LF) heat-flux method, preserves its method-level validation figures, and adds representative outputs from the reduced Majeski–Kunz–Squire (2024, hereafter MKS24) analysis workflow.

The method-level results show that the implemented update follows the intended finite-volume variable strategy and passes checks including a one-dimensional oblique linear initial-value problem using the Squire et al. Figure 17 background state. The reduced nonlinear figures exercise analysis and plotting paths only: they are not standard-resolution, steady-state, or quantitative MKS24 reproduction results. The current execution and reference-admission status is maintained in `docs/cgl_lf_mks24_reproduction_implementation_plan.md` and in the Sphinx CGL–LF validation guide.

2 Conserved Variables

The hyperbolic CGL system evolves total energy and the conserved anisotropy variable

$$A = \rho \log \left[\frac{p_{\perp}}{p_{\parallel}} \frac{\rho^2}{|B|^3} \right]. \quad (1)$$

The legacy array slot is still named `IMU`, with `IAN` used as an alias in CGL code. The Riemann solve remains in `A`, which avoids the small- $|B|$ pathologies associated with evolving magnetic moment directly.

The LF STS operator temporarily changes only the interpretation of the `IAN` slot:

$$A \rightarrow \mu_B = \frac{p_{\perp}}{|B|}. \quad (2)$$

The standard conserved-variable STS update then advances (E, μ_B) with the LF flux divergence. Immediately after the RKL2 stage, the EOS converts $\mu_B \rightarrow A$, and the normal CGL conserved-to-primitive recovery restores $(p_{\parallel}, p_{\perp})$.

In the current source tree the LF parabolic operator is owned by `src/diffusion/cgl_landau_fluid.*`; the CGL state and constraint logic are implemented in `src/eos/cgl_mhd.cpp` and `src/eos/cgl_physics.hpp`. The quantitative method tests use `src/pgen/tests/cgl_landau_fluid.cpp`, while the reduced MKS24 initialization and history diagnostics use `src/pgen/tests/cgl_lf_paper.cpp`.

3 Landau-Fluid Closure

The implemented closure uses the Snyder–Hammett–Dorland 3+1 heat fluxes,

$$q_{\parallel} = -\chi_{\parallel} \rho \nabla_{\parallel} \left(\frac{p_{\parallel}}{\rho} \right), \quad (3)$$

$$q_{\perp} = -\chi_{\perp} \left[\rho \nabla_{\parallel} \left(\frac{p_{\perp}}{\rho} \right) - p_{\perp} \left(1 - \frac{p_{\perp}}{p_{\parallel}} \right) \frac{\nabla_{\parallel} |B|}{|B|} \right], \quad (4)$$

with the finite-collision Squire et al. coefficients

$$\chi_{\perp} = \frac{2c_{\parallel}^2}{\sqrt{2\pi} c_{\parallel} |k_{\parallel}| + \nu_{\text{eff}}}, \quad \chi_{\parallel} = \frac{8c_{\parallel}^2}{\sqrt{8\pi} c_{\parallel} |k_{\parallel}| + (3\pi - 8)\nu_{\text{eff}}}. \quad (5)$$

Here ν_{eff} is the background collision frequency plus any active mirror/firehose limiter scattering at the face. In the collisionless limit, these reduce to $\chi_{\perp} = \sqrt{2/\pi} c_{\parallel}/|k_{\parallel}|$ and $\chi_{\parallel} = \sqrt{8/\pi} c_{\parallel}/|k_{\parallel}|$. Here `lf_k_parallel` is the closure wavenumber $|k_{\parallel}|$, not a conductivity. In local mode, $c_{\parallel} = \sqrt{p_{\parallel}/\rho}$ is computed from the live face state. In background mode, only this prefactor is frozen to `lf_c_parallel0`; all gradients and thermodynamic variables remain live.

The face fluxes coupled into STS are

$$F_E^i = \hat{b}^i \left(q_{\perp} + \frac{1}{2} q_{\parallel} \right), \quad F_{\mu_B}^i = \hat{b}^i \frac{q_{\perp}}{|B|}. \quad (6)$$

4 Heat-Flux Cap

Following Squire et al. (2023), equations 3.1–3.2, the unlimited heat flux q_L is capped as

$$q = \frac{q_L q_{\text{max}}}{q_{\text{max}} + |q_L|}. \quad (7)$$

The implemented caps are

$$q_{\perp, \text{max}} = \sqrt{\frac{2}{\pi}} c_{\parallel} p_{\perp}, \quad q_{\parallel, \text{max}} = \sqrt{\frac{8}{\pi}} c_{\parallel} p_{\parallel}. \quad (8)$$

The cap is applied after computing the unlimited flux with the collision-aware coefficient, so collisions and limiter scattering suppress the flux before the smooth free-streaming bound is applied.

5 Threshold Policy and Retained Diagnostics

The current implementation separates backward-compatible operator tests from MKS24 production definitions. Existing method checks retain the default oblique-firehose policy. MKS24 decks set `cgl_firehose_threshold = parallel`, selecting $\beta\Delta \leq -2$, together with the mirror threshold $\beta\Delta \geq 1$. Hard-wall MKS24 decks additionally set `limiter_hardwall = true`; newly unstable primitive states are then projected to the selected threshold while preserving $p_{\perp} + p_{\parallel}/2$, rather than relying on finite-rate limiter relaxation to approximate an algebraic wall.

LF-enabled runs append persistent history counters for STS stages, floors, non-finite and non-positive states, ordinary limiter occupancy, emergency hard-bound crossings, heat-flux-cap activity, applied heat-flux contractions, and hard-wall projection applications. When enabled for paper decks, `cgl_lf_record_pressure_work` also retains explicit-RK-applied total and anisotropic CGL pressure-traction work. Strict validation rejects floors, invalid states, or emergency-bound crossings; hard-wall projection activity is an expected model action, not by itself a safety failure.

6 Validation Workflow

Generate the method-validation figures from the repository root with

```
python3 scripts/cgl_lf_workflow.py full --output-dir /path/to/full-bundle
```

or with the compatibility command `scripts/run_cgl_lf_validation.sh`. The driver writes an ignored result bundle containing a manifest, readable summary, logs, histories, machine-readable validation CSV files, and figures. The five method figures included below were regenerated from the retained passing bundle `build-cgl-implementation/cgl_lf_runs/20260524-final-full-v2`.

Each test prints the measured value, reference value or error norm, tolerance, and pass/fail status. The plotted points are read from the same diagnostic CSV files produced by the passing runs.

Generate reduced nonlinear diagnostic products with

```
python3 scripts/cgl_lf_workflow.py paper-convergence
python3 scripts/cgl_lf_workflow.py paper-analyze \
  --output-dir /path/to/paper-convergence-bundle
```

The representative nonlinear figures below were regenerated with the current analyzer from the retained passing `20260525-paper-convergence-pressure-work-v1` bundle. Its cases run only through $t = 0.02$ on reduced grids and therefore test product plumbing, not statistically saturated turbulence or paper agreement.

7 Validation Problems and Tolerances

8 Quantitative Results

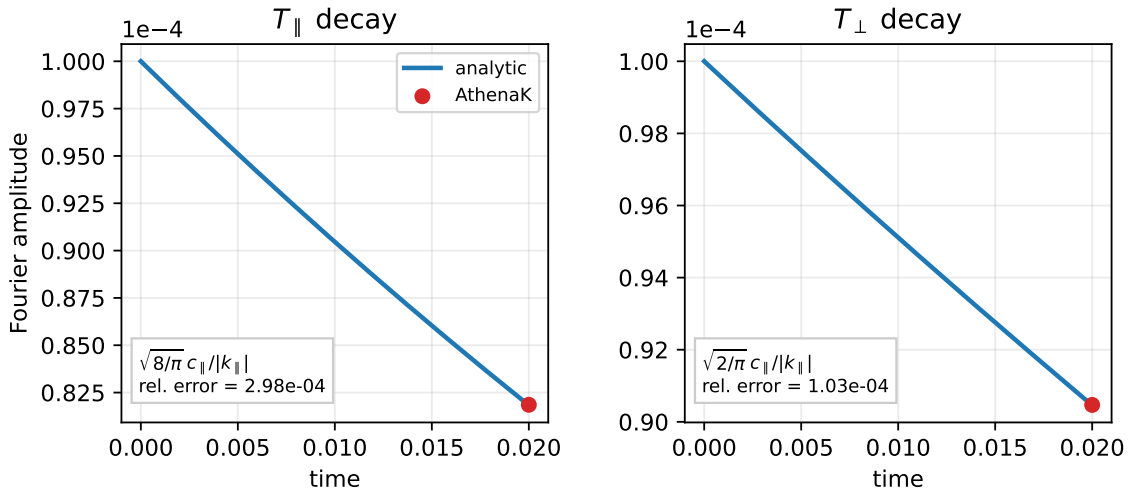


Figure 1: Constant-coefficient decay tests. The $T_{\parallel}$ case uses $\chi_{\parallel} = \sqrt{8/\pi} c_{\parallel} / |k_{\parallel}|$; the $T_{\perp}$ case uses $\chi_{\perp} = \sqrt{2/\pi} c_{\parallel} / |k_{\parallel}|$. The red point is the final AthenaK Fourier amplitude and the blue curve is the analytic constant-coefficient decay.

Input	Purpose	Assertion
cgl_transform_check	$A \leftrightarrow \mu_B$ and primitive recovery	pass
cgl_lf_quant_parallel	sinusoidal $T_{\parallel}$ decay	final Fourier amplitude within 5% of analytic decay
cgl_lf_quant_parallel_collisional	sinusoidal $T_{\parallel}$ decay with $\nu_c > 0$	final Fourier amplitude within 7% of finite-collision analytic decay
cgl_lf_quant_perp	sinusoidal $T_{\perp}$ decay	final Fourier amplitude within 5% of analytic decay
cgl_lf_quant_perp_collisional	sinusoidal $T_{\perp}$ decay with $\nu_c > 0$	final Fourier amplitude within 7% of finite-collision analytic decay
cgl_lf_quant_grad_b	isolated $\nabla_{\parallel} B $ term	RMS relative error below 10%, positive response correlation
cgl_lf_flux_limiter	equation 3.2 heat-flux cap	$ q \leq q_{\max}$, sign preserved, update within 20%
cgl_lf_limiter_heat_flux_suppression	limiter-collision suppression of heat fluxes	active limiter scattering reduces the unlimited LF flux by more than 20x
cgl_lf_limiter_mirror	mirror-biased admissibility stress	intermediate STS states finite, positive, and backup bounded
cgl_lf_limiter_firehose	firehose-biased admissibility stress	intermediate STS states finite, positive, and backup bounded
cgl_lf_field_wave	field-aligned LF wave reference	complex Fourier mode within 25% of offline reference
cgl_lf_paper_oblique_wave	Figure 17 background, LF enabled	guard tolerance 40%; observed errors $< 3 \times 10^{-6}$
cgl_pure_paper_oblique_wave	Figure 17 background, no LF	guard tolerance 40%; observed errors $< 3 \times 10^{-6}$
cgl_pure_paper_eigen_*	exact pure-CGL Alfvén/slow/fast modes	all tracked Fourier components within 7.5%
cgl_lf_paper_eigen_*	exact CGL-LF Alfvén/slow/fast modes	all tracked Fourier components within 7.5%

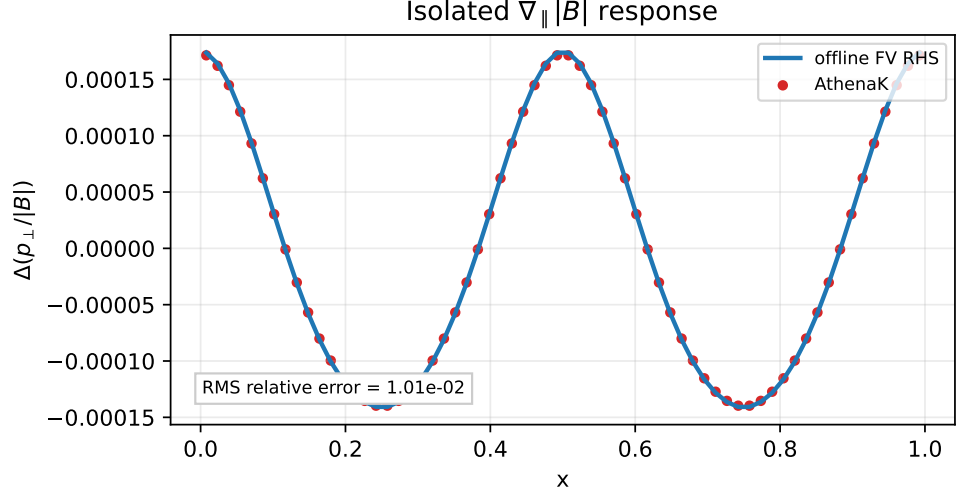


Figure 2: Isolated $\nabla_{||}|B|$ term in the perpendicular heat flux. The initial thermal gradients are zero; the plotted response is the short-time change in $p_{\perp}/|B|$ compared with an offline finite-volume reconstruction of the initial LF RHS.

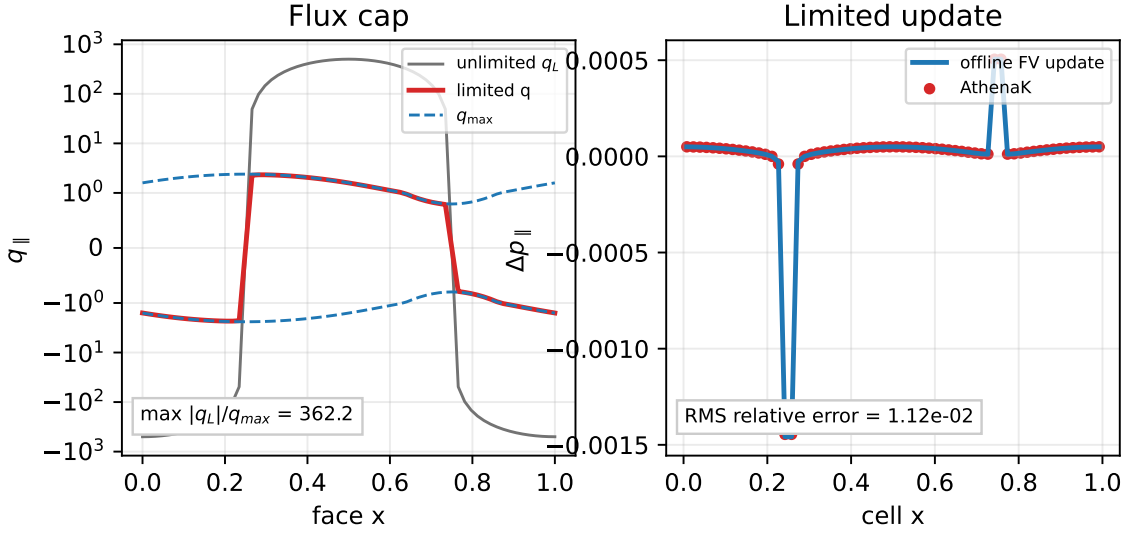


Figure 3: Heat-flux limiter test. The left panel shows the unlimited $q_{||,L}$, the capped $q_{||}$, and $\pm q_{||,max}$. The right panel checks that the pressure update produced by the limited flux matches the offline finite-volume update.

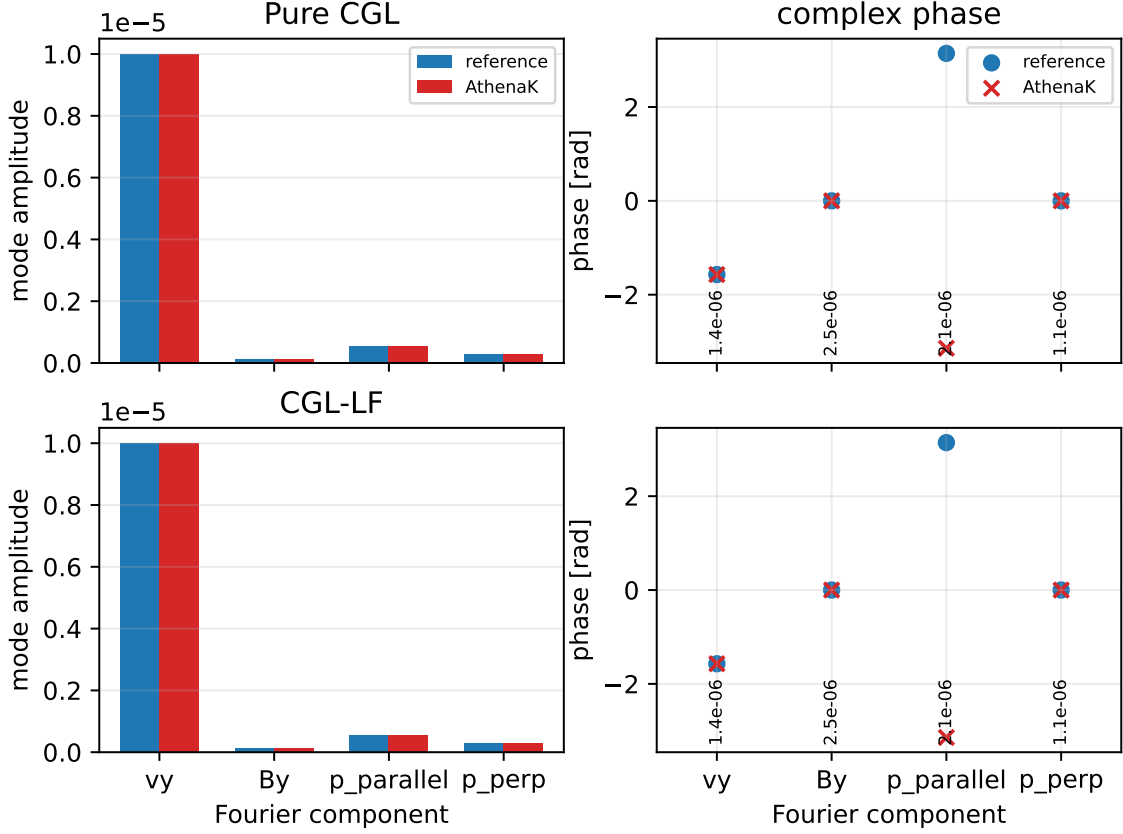


Figure 4: Figure 17-style oblique linear initial-value check. The plotted quantities are the final complex Fourier amplitudes of u_y , B_y , $p_{\parallel}$, and $p_{\perp}$ for the pure-CGL and CGL-LF inputs, compared with the offline linearized initial-value reference used by the quantitative pgen. The vertical phase-panel labels are per-variable relative errors.

9 Figure 17-Style Linear Initial-Value Check

The paper-style inputs use the Figure 17 background

$$\rho_0 = 1, \quad p_{\parallel 0} = p_{\perp 0} = 5, \quad u_0 = 0, \quad B_0 = (1, \sqrt{2}, 0.5), \quad k = 2\pi\hat{x}, \quad kL = |k|. \quad (9)$$

The test initializes a small transverse velocity perturbation, not a CGL eigenmode, and compares the final complex Fourier amplitudes of u_y , B_y , $p_{\parallel}$, and $p_{\perp}$ against an offline linearized CGL–LF initial-value reference integrated with the same background state and closure coefficients. Two inputs are provided: one with the LF heat flux enabled and one with pure CGL. This checks the same method class as the Appendix A.2/Figure 17 linear-wave convergence problem.

10 Figure 17-Style Eigenmode Checks

The validation suite now also includes exact eigenmode initializations for the Alfvén, slow-like, and fast-like positive-frequency branches of the same linearized Figure 17 system. The script `scripts/generate_cgl_lf_eigenmode_inputs.py` constructs the linear CGL or CGL–LF matrix for the AthenaK variables $(\rho, u_x, u_y, u_z, B_y, B_z, p_{\parallel}, p_{\perp})$, diagonalizes it, normalizes the selected eigenvectors, and writes their complex amplitudes and eigenvalues into AthenaK input files. The pgen initializes $\text{Re}[\delta U_k \exp(ikx)]$ and compares the final Fourier mode with $\delta U_k \exp(\lambda t)$, where λ is the supplied eigenvalue. These inputs are the direct replacement for using a single transverse-velocity seed as a proxy for the Squire/Jono linear-wave comparisons.

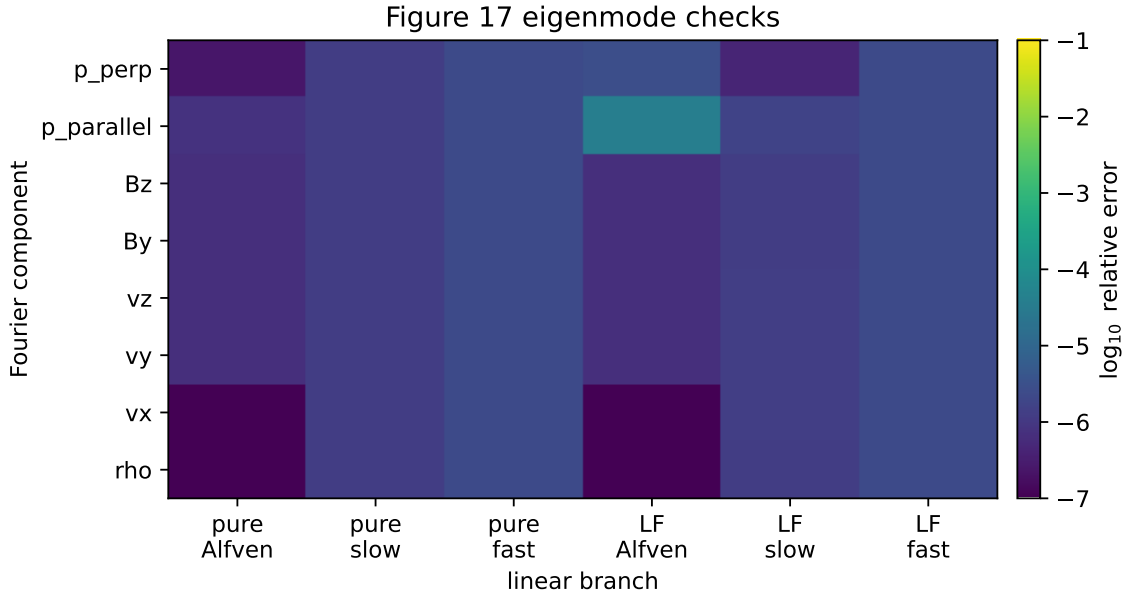


Figure 5: Exact eigenmode checks on the Figure 17 background. Each column is a pure-CGL or CGL–LF Alfvén, slow-like, or fast-like positive-frequency branch. The color scale shows the complex Fourier relative error in each primitive or magnetic component after a short AthenaK evolution compared with the supplied linear eigenmode prediction $\delta U_k \exp(\lambda t)$. The white numbers give the maximum tracked-component error in each branch.

11 Reduced MKS24-Oriented Analysis Products

The `paper-convergence` workflow supplies short, reduced forced turbulence bundles used to validate nonlinear diagnostics and rendering. The figures in this section use the current analyzer on retained snapshots from `20260525-paper-convergence-pressure-work-v1`. The bundle contains three resolution-path cases and heat-flux/policy variants, all with clean LF safety diagnostics. Because these cases terminate at $t = 0.02$, their curves must not be interpreted as steady turbulent spectra or MKS24 comparisons.

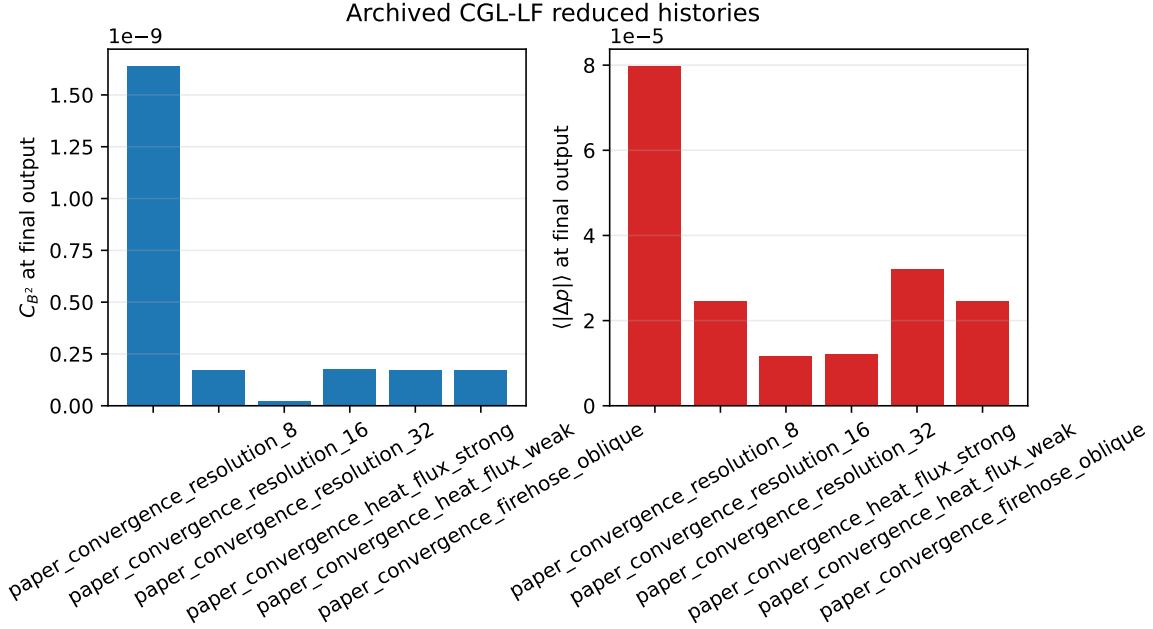


Figure 6: Reduced history-product rendering for the nonlinear analysis path. The plotted global summaries verify ingestion of archived history data across the reduced case matrix; they do not establish converged physical trends.

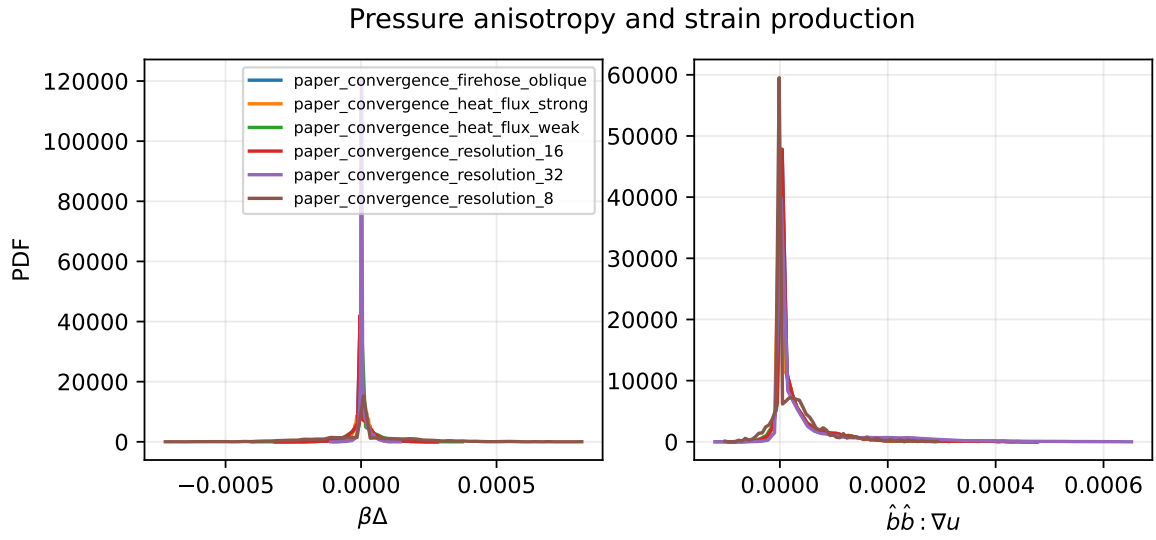


Figure 7: Reduced pressure-anisotropy and local-strain PDF rendering. These products expose $\beta\Delta$ and $\hat{\mathbf{b}}\hat{\mathbf{b}} : \nabla \mathbf{u}$ for later panel comparisons once authorized statistically useful runs exist.

Pressure-density joint PDF: paper_convergence_resolution_32

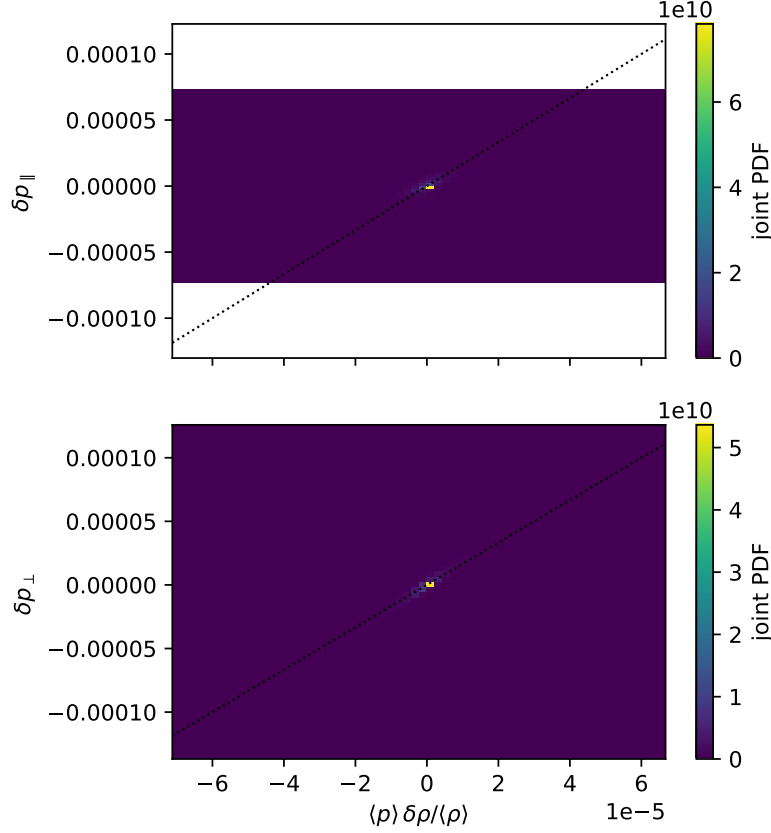


Figure 8: Representative Figure 2(a)-coordinate joint-PDF diagnostic from the $32 \times 32 \times 64$ reduced case. The analyzer now emits $\langle p \rangle \delta \rho / \langle \rho \rangle$ versus $\delta p_{||}$ and $\delta p_{\perp}$. The admitted digitized MKS24 surface route is separate from this reduced simulation output and requires a matching standard case.

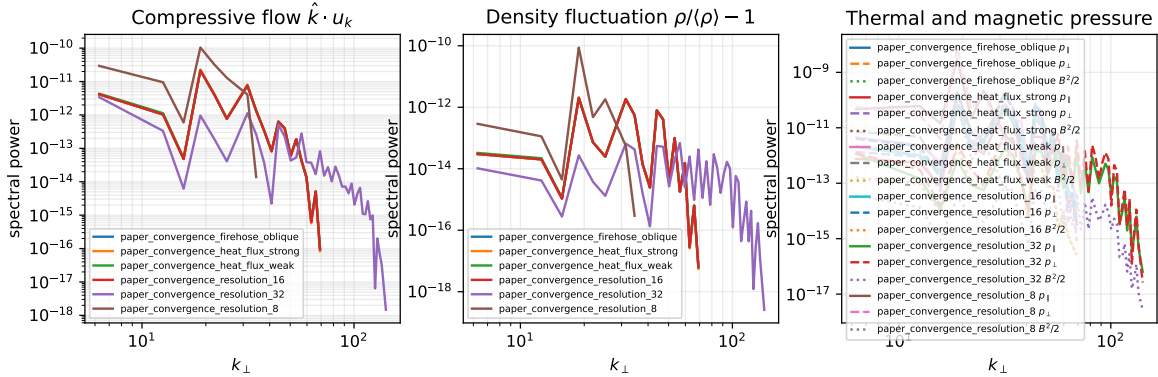


Figure 9: Current reduced rendering of the Figure 3/4(b)/6(a)-oriented spectral fields: compressive velocity projection, normalized density fluctuation, and thermal/magnetic-pressure products. Their plotted MKS24 absolute ordinates remain excluded from reference comparison until an external discrete Fourier-normalization mapping or original numeric data are available.

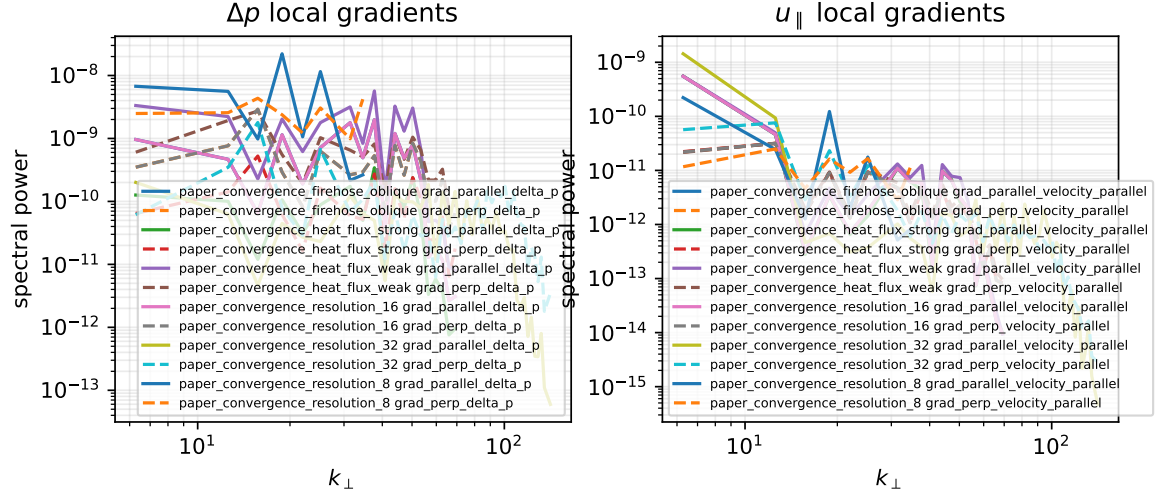


Figure 10: Projected pressure-anisotropy and velocity-gradient spectrum products from the reduced bundle. These are diagnostic-product checks and not admitted absolute-ordinate paper comparisons.

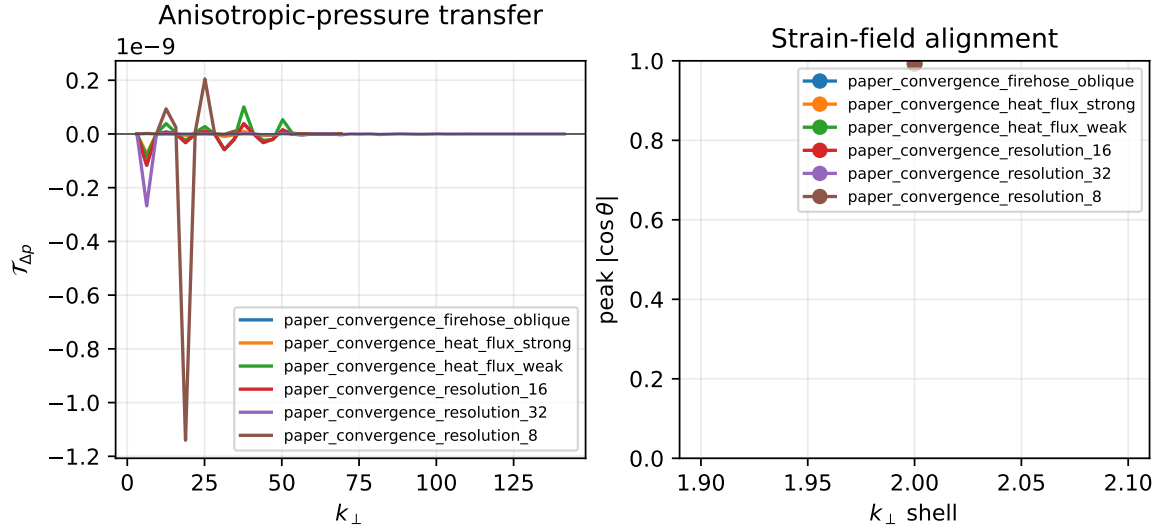


Figure 11: Pressure-stress transfer and alignment-product rendering. The analyzer supports the dimensionless MKS24-normalized transfer comparison and selected-shell or peak-alignment comparisons when supplied with provenance-qualified references and matching executed cases.

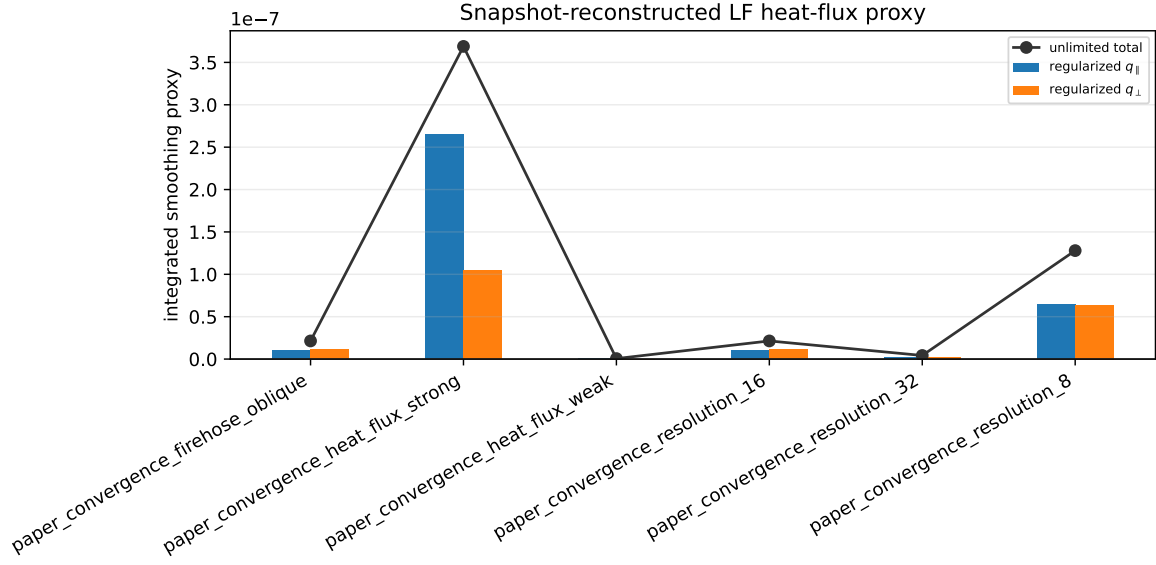


Figure 12: Snapshot-reconstructed heat-flux smoothing proxy for reduced cases. It is useful for interpretation of retained snapshots, but is not the applied finite-volume heat-flux contraction or a total energy closure.

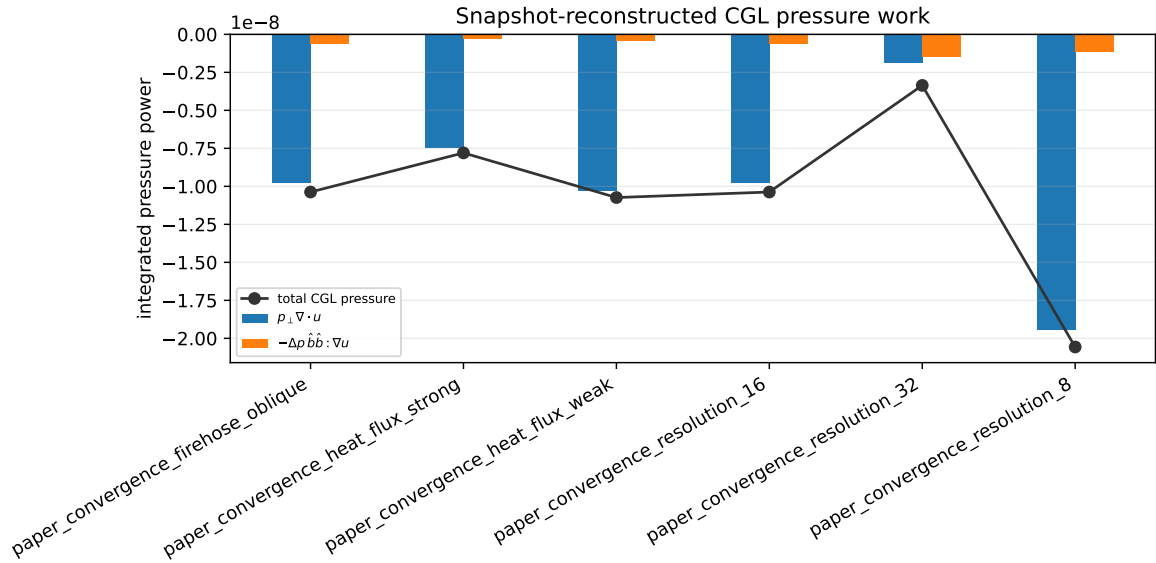


Figure 13: Snapshot-derived CGL pressure-work decomposition, including the anisotropic stress contribution checked against direct pressure transfer. Paper decks also retain applied pressure-traction ledger histories; neither the reduced snapshot plot nor an unevaluated standard deck establishes a production result.

12 MKS24 Reproduction Boundary

The current branch defines guarded standard, limiter-frequency, heat-flux, compressive, and scale-separation workflows and supports checksum/uncertainty-qualified external reference comparisons. The admitted published reference products presently include Figure 2(a) transformed sampled surfaces, Figure 2(b) histories, normalized Figure 4(a) PDFs, normalized Figure 5(b) eddy scales, dimensionless Figure 7 transfer ratios, Figure 8 selected-shell alignment PDFs, Figure 9 and Figure 11/12 alignment curves, and Figure 13(b),(d) dimensionless products.

Remaining absolute spectral or strain ordinate panels are intentionally not admitted: the paper and cited donor source define shell spectra and plotted units but do not state the discrete Fourier normalization needed to map absolute digitized ordinates into AthenaK products. Those panels require author/archive numeric data, donor diagnostic code, or an explicit normalization statement. Production executions and quantitative MKS24 simulation-to-reference comparisons additionally require explicit authorization; none is claimed by this note.

13 Build and Runtime Status

The imported method figures come from a passing local `full` workflow bundle. The reduced nonlinear figures come from a passing `paper-convergence` bundle reanalyzed with the current plotting code. Warning-strict Sphinx documentation and this L^AT_EX build are checked when the report is updated. Production/statistical and paper-reference comparison gates remain outside this method-document build.

References

- [1] J. Squire et al., *Pressure anisotropy and viscous heating in weakly collisional plasma turbulence*, arXiv:2303.00468, 2023.
- [2] S. Majeski, M. W. Kunz, and J. Squire, *Self-organization in collisionless, high- β turbulence*, arXiv:2405.02418, 2024.